STAFF REVIEW · H446 316 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Make a reasoned judgement, with relevant supporting evidence.

2. A relevant limitation to the preferred choice.

3. It decides which advantages and limitations matter.

4. Sum0 from XOR; carry1 from AND.

I Do — source worked steps

1. Explain the worked method — A strong answer explains mechanical access, portability and the need for backups.

We Do — guided CHECK

1. Explain the worked method
Match security, home connectivity and backup provision to the school context.

You Do — source mark schemes

1. Multi-device access and shared controls [2]; outage/privacy limitation [2]; justified view based on school connectivity/backup [2].

2. Indicative content: GPU parallel throughput, CPU suitability for serial/branch-heavy operations, transfer overhead, cost and energy. Practice marking (authored guidance, not an OCR mark scheme): 7–9 = accurate, sustained evaluation of both options, applied to this scenario with a supported judgement; 4–6 = relevant comparison with some development and a partly justified decision; 1–3 = basic statements, limited context or unsupported decision; 0 = no creditworthy response. Use best-fit and credit equivalent valid arguments.
Model evaluation: For a large batch of independent video frames, a GPU can perform many similar pixel operations simultaneously, so it may substantially improve throughput. A multicore CPU is more flexible for the control logic, branching and varied tasks used elsewhere in the application. Moving data between CPU and GPU memory can add latency, so a small job may finish faster without GPU offloading. Power consumption, hardware cost and development effort also matter. For large-scale video analysis I would use GPU acceleration for suitable parallel stages, retaining CPU cores for orchestration and general tasks. The decision should be tested on realistic workloads rather than inferred from core counts alone.

3. Indicative content: recovery point objective, local-site failure/ransomware, backup frequency, off-site independence, encryption, access controls, restore tests and cost. Practice marking (authored guidance, not an OCR mark scheme): 7–9 = accurate, sustained evaluation of both options, applied to this scenario with a supported judgement; 4–6 = relevant comparison with some development and a partly justified decision; 1–3 = basic statements, limited context or unsupported decision; 0 = no creditworthy response. Use best-fit and credit equivalent valid arguments.

Model evaluation: A nightly local backup is inexpensive and may restore an accidentally deleted file quickly, but a failure just before the next backup could lose almost a day's work. Fire, theft or ransomware could also affect both live files and local copies. Secure, independently controlled off-site backups taken more frequently reduce the likely loss of recent records and protect against a building-level incident. However, they bring ongoing cost, connectivity and data-protection responsibilities. I would recommend frequent encrypted off-site or immutable backups alongside a convenient local copy, with restricted access and regular restore tests. The school should choose frequency against an agreed maximum acceptable data loss and recovery time.

Exit Ticket — source answer

1. False. Relevant explanation and contextual evaluation matter.